

created, the Spline IK curve is like any other curve and can be animated by using clusters or blend shapes.

6.4 Constraints

Constraints are a way to automatically control an object's position, scale, or orientation. Constraints are used in animation as well as in character rigging to provide animators with ways to attach parts of a character's body to other objects or parts of a scene.

6.4.1 Position

In animation, a Position constraint causes one object to move to and follow the position of another object, or the average position of several objects. Position constraints are particularly useful when you want an object to match another object's position while keeping it outside the hierarchy, such as when a character is lifting something. When the character lifts the ball, the ball is constrained to the palm of the hand. This allows the ball to move with the hand as the hand itself is animated.

6.4.2 LookAt

A LookAt constraint constrains an object's orientation so that the object aims at other objects. In character setup, a typical use of a LookAt constraint is to set up a locator that controls eyeball movement. The eyeball is aim constrained to the locator and rotates to follow it.

6.4.3 Orientation

An Orientation constraint matches one object's rotation to another. This can be used in character animation to set up control objects that affect other objects, such as bones within a rig.

The bone controlled by an Orientation constraint driven by the circle. Rotating the circle swivels the bone.

6.4.4 Link

A Link constraint allows you to animate the links in a hierarchy. Just as with hierarchical linking, a Link constraint matches the relative position, rotation, and scale of one object to another. The Link constraint, however, can be animated to change the link to a different object at a specific frame. This allows for characters to set down and pick up objects. The Link constraint allows the ball to be linked to one hand and then another as the ball passes between them.